# ASSIGNMENT-5

K-Means is an **unsupervised machine learning algorithm** used to group data into **K clusters** based on similarity.

- Choose number of clusters **K**
- Assign points to nearest cluster center (centroid)
- Update centroids
- Repeat until stable

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### 🔹 Simple Example

Suppose you have points like:

(1,2), (2,3), (8,9), (9,10)

If **K = 2**, algorithm groups them into:

- Cluster 1 → near (1,2)
- Cluster 2 → near (8,9)

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[1]

```python
import numpy as np

data = np.array([
    [1, 2],
    [2, 3],
    [8, 9],
    [9, 10]
])

k = 2
centroids = data[:k]

for _ in range(10):
    clusters = [[] for _ in range(k)]

    for point in data:
        distances = [np.linalg.norm(point - c) for c in centroids]
        cluster_index = np.argmin(distances)
        clusters[cluster_index].append(point)

    new_centroids = []
    for cluster in clusters:
        new_centroids.append(np.mean(cluster, axis=0))

    new_centroids = np.array(new_centroids)
```

Release notes

Please follow our blog to see more info
tricks, and featured notebooks such as

2026-02-23

- Enterprise Runtimes are now ava
notebook you love with the enter
  - Persistent runtimes
  - Configurable hardware & p
  - Seamless Vertex AI & BigQ
  - Scale from idea to produc
- Launched official PyCharm Colal
- Published Colab's CPU Docker ru
docker.pkg.dev/colab-images/pu
- Python package upgrades
  - bigframes 2.31.0 -> 2.35.0
  - bokeh 3.7.3 -> 3.8.2
  - cudf-cu12 25.10.0 -> 26.2.
  - cudf-polars-cu12 25.10.0
  - cuml-cu12 25.10.0 -> 26.2.
  - cupy-cuda12x 13.6.0 -> 14
  - dask 2025.9.1 -> 2026.1.2
  - dask-cuda 25.10.0 -> 26.2
  - dask-cudf-cu12 25.10.0 ->
  - fastapi: 0.123.10 -> 0.133.0
  - google-adk 1.21.0 -> 1.25.1

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[1]

```python
        new_centroids.append(np.mean(cluster, axis=0))

    new_centroids = np.array(new_centroids)

    if np.all(centroids == new_centroids):
        break

    centroids = new_centroids

for i, cluster in enumerate(clusters):
    print(f"Cluster {i+1}: {cluster}")

print("Centroids:", centroids)
```

```
Cluster 1: [array([1, 2]), array([2, 3])]
Cluster 2: [array([8, 9]), array([ 9, 10])]
Centroids: [[1.5 2.5]
 [8.5 9.5]]
```

Releas

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- Py